VIT Vellore Institute of Technology (Deemed to be University under section 3 of UGC Act, 1956) Invention Disclosure Format (IDF)-B	Document No.	
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1. Title of the invention:

A Wearable Assistive Navigation Apparatus and Method for Real-Time Voice-Guided and Haptic-Feedback Object Avoidance Using Proximity-Weighted Threat Prioritization Algorithm with Adaptive Announcement Scheduling for Visually Impaired Persons

2. Field / Area of invention

- This invention relates to the field of wearable assistive technology, specifically to edge-computing wearable apparatus integrated with deep learning-based object detection, inertial motion sensing, and multi-modal output actuators for real-time navigation guidance for visually impaired persons.
- The invention pertains to a closed-loop wearable cyber-physical apparatus that applies a novel Proximity-Weighted Threat Prioritization Algorithm (PWTPA), Adaptive Announcement Scheduling Engine (AASE), and Class-Split Safety Architecture (CSSA) to detect indoor objects, compute mathematically prioritized threat scores, and deliver synchronized voice, haptic, and spatial audio guidance to visually impaired users through integrated hardware actuators.
- The scope of this invention explicitly excludes outdoor GPS-based navigation systems, population-level accessibility advisory platforms, image captioning systems without real-time obstacle warning capability, and any application that does not involve simultaneous multi-modal output delivery through integrated wearable hardware with formal threat prioritization algorithms.

3. Prior patents and publications from literature

The following representative prior art was reviewed to map gaps relevant to the present invention:

Patent / Publication / Year	Core Contribution	Data / Method Focus	Limitations / Gaps Identified
US10299982B2(Patent, 2019)	3D environment reconstruction for blind navigation using mobile camera with touch-screen virtual exploration	Mobile camera + MVS/SfM 3D reconstruction; touch-screen environment exploration	Focuses on 3D spatial reconstruction and touch exploration rather than real-time object-level threat prioritization with PWTPA scoring, announcement scheduling, or multi-modal haptic/voice output
US9792501B1(P atent, 2017)	Wearable computer vision device (DVIA) using video sensors with haptic transducers for navigational object reporting	Wearable camera + machine vision; haptic transducer feedback for detected objects	Provides haptic feedback for detected objects but lacks formal threat prioritization algorithm, announcement scheduling discipline, IMU motion compensation, class-split safety architecture, or bone conduction audio

Patent / Publication / Year	Core Contribution	Data / Method Focus	Limitations / Gaps Identified
US10528815B2(Patent, 2020)	Continuation of DVIA wearable assistive device with enhanced haptic feedback for visually impaired navigation	Wearable video + haptic; enhanced object identification and reporting	Extension of US9792501B1 with similar limitations: no mathematical threat scoring framework, no announcement cooldown/escalation logic, no per-class proximity calibration
US10371544B2(Patent, 2019)	Vibrating haptic navigation device for blind and visually impaired using directional vibration cues	Wearable haptic band; vibration-based directional navigation cues	Provides directional haptic cues but relies on external positioning (GPS/beacons) rather than camera-based object detection; no deep learning, no threat scoring, no announcement scheduling
US9801778B2(P atent, 2017)	System for alerting visually impaired users of nearby objects using collision frustum and vibrotactile feedback	Depth sensing + collision frustum detection; vibrotactile belt alerts	Uses collision frustum geometry for proximity alerts but lacks object classification capability, per-class proximity thresholds, escalation override, or multi-modal synchronized output
US9613505B2(P atent, 2016)	Localized extremity guidance system using vibration interface to navigate user hands to detected objects	Vibration interface on extremity; sensor-based object trajectory guidance	Guides user extremities to objects rather than providing walking navigation; no obstacle threat prioritization, announcement discipline, or voice/sonar output modalities
US10690771B2(Patent, 2020)	Object detection using human echolocation principles with transducer arrays for visually impaired	Echolocation transducer array; reflected signal processing for obstacle detection	Relies on acoustic echo-based detection rather than camera-based deep learning; no object classification, directional zone analysis, or adaptive announcement scheduling
Shen et al.(Sensors, 2022)	Wearable assistive device for blind pedestrians using compressed YOLOv3 and SMA tactile actuators	Compressed YOLOv3 on mobile device; SMA wire tactile display for object proximity	Provides tactile proximity feedback but lacks formal threat prioritization formula, announcement scheduling logic, bone conduction audio, IMU motion compensation, or class-split safety architecture
Wang et al.(Sensors, 2024)	YOLO-OD obstacle detection method for visually impaired navigation with feature weighting blocks	Modified YOLO with FWB, ABB, EFAH modules; outdoor obstacle detection benchmarks	Proposes detection architecture improvements but does not address announcement management, multi-modal output delivery, threat prioritization algorithms, or wearable hardware integration
Bouteraa et al.(Micromachine s, 2021)	Wearable assistive device integrating fuzzy decision support with mixed haptic-voice interface for BVIP	Ultrasonic sensors + fuzzy logic DSS; BeagleBone Black + ROS; haptic-voice output	Uses ultrasonic sensors and fuzzy logic rather than camera-based deep learning; no per-class object classification, no formal announcement scheduling, no IMU-assisted detection stabilization
Automation MDPI(Journal, 2026)	Vision-based smart wearable navigation using fine-tuned YOLOv5n with depth camera and voice feedback	YOLOv5n + depth camera; voice feedback for detected obstacles with distance	Provides voice feedback with distance estimation but lacks threat prioritization scoring, announcement cooldown/escalation discipline, haptic feedback, or class-split safety architecture
Bai et al.(arXiv, 2019)	Wearable travel aid with ground detection algorithm and 2.5D object detection for visually impaired	RGB-D camera; ground detection + walkable direction search; audio feedback	Focuses on ground-level navigation and walkable path detection rather than object-level threat prioritization with PWTPA scoring, multi-modal output, or announcement scheduling

4. Summary and background of the invention

According to the World Health Organization (WHO, 2023), at least 2.2 billion people globally have a near or distance vision impairment, with an estimated 43 million classified as blind and 295 million experiencing moderate-to-severe vision impairment (Lancet Global Health, GBD 2020), with indoor navigation posing a clinically unresolved challenge. Indoor environments contain dynamic obstacle configurations — furniture, household objects, doorways, and people — that present continuous collision risks. Conventional assistive tools (white canes, guide dogs) provide limited spatial awareness restricted to immediate contact and cannot identify specific objects, communicate their direction, or adapt guidance based on relative threat priority.

Existing electronic assistive solutions fall into two categories, each with critical limitations. Hardware-sensor systems (ultrasonic, infrared) detect obstacles at limited ranges but cannot classify object types, provide directional guidance, or differentiate threat levels. Deep learning software systems on smartphones can classify objects but produce excessive announcements (every detected object spoken aloud), lack formal mathematical frameworks for determining announcement priority, and do not provide multi-modal feedback (voice + haptic + spatial audio) to accommodate varying noise environments. No existing system combines wearable hardware with integrated edge computing, formal threat prioritization algorithms, and multi-modal actuator output.

The present invention discloses a wearable assistive navigation apparatus that bridges this gap by embedding a wide-angle camera, 6-axis IMU, Edge TPU neural accelerator, bone conduction speaker, and 3-point vibrotactile haptic array within a chest-mounted wearable unit. The apparatus implements three novel algorithmic contributions: (1) the PWTPA that computes $T(o) = \alpha \cdot P(o) + \beta \cdot C(o) + \gamma \cdot S(o) + \delta \cdot M(o)$ for each detected object; (2) the AASE with four-stage filtering (persistence, cooldown, gap, escalation override) achieving 95.6% announcement reduction; and (3) the CSSA preventing misclassification-induced false warnings. The apparatus produces tangible physical effects through bone conduction (sound via bone vibration), haptic motors (tactile direction/proximity encoding), and IMU-assisted detection stabilization (38.4% fewer missed detections). It is architecturally and functionally distinct from all cited prior art.

5. Objective(s) of the invention

- To provide a wearable navigation apparatus integrating camera, IMU, Edge TPU, bone conduction speaker, and haptic motors that computes PWTPA threat scores $T(o) = \alpha \cdot P(o) + \beta \cdot C(o) + \gamma \cdot S(o) + \delta \cdot M(o)$ for each detected object and delivers prioritized multi-modal guidance.
- To enable an Adaptive Announcement Scheduling Engine (AASE) with four-stage filtering (persistence, cooldown, gap, escalation override) that reduces announcement frequency by 95.6% compared to naïve per-frame systems while maintaining zero missed critical warnings for approaching obstacles.
- To implement a Class-Split Safety Architecture (CSSA) that categorically separates obstacle classes from findable classes, preventing deep learning model misclassification from producing false obstacle warnings — a safety-by-design mechanism absent from all cited prior art.

- To validate an IMU-assisted adaptive confidence threshold mechanism that dynamically adjusts detection sensitivity based on real-time motion magnitude, reducing missed detections during walking by 38.4%.
- To deliver synchronized multi-modal output through bone conduction voice guidance (preserving ambient hearing), spatial sonar audio (stereo-panned proximity-modulated beeps), and vibrotactile haptic patterns (direction and proximity encoded through 3-point motor activation and pulse frequency modulation).

5A. Novelty and Key Advantage of the Invention

The present invention establishes a dedicated wearable assistive navigation apparatus and algorithmic framework that is architecturally distinct from all previously developed assistive systems. Unlike prior systems that concentrate on object classification, scene description, or binary proximity alerts, this invention is limited to a specifically defined hardware-integrated optimization and enforcement workflow.

The PWTPA algorithm, AASE scheduling engine, CSSA safety architecture, IMU-assisted motion compensation, and multi-modal actuator delivery layer are all specifically designed for and confined to this invention. A significant benefit of this invention is its clear architectural separation:

- To the best of the inventors' knowledge, it does not reuse, extend, or rely on any workflows, system components, or analytical outputs claimed in earlier assistive technology patents.
- It prevents functional overlap by operating with distinct hardware sensor inputs (wide-angle camera + 6-axis IMU), independent edge-computing processing (Edge TPU), and non-overlapping multi-modal physical enforcement outputs (bone conduction + haptic motors + sonar audio).
- It introduces a self-sufficient closed-loop detection-analysis-prioritization-scheduling-delivery workflow, ensuring that no previously patented advisory or detection logic is invoked.

This architectural separation is believed to reinforce strong novelty and support independent defensibility, as the invention can be implemented, deployed, and assessed without dependency on any prior assistive technology filings. Final determination of novelty and freedom-to-operate is subject to formal search and examination.

Identification of Closest Prior Art: Among all cited references, US9801778B2 (collision frustum vibrotactile system) represents the functionally closest prior art, as it combines depth-based proximity detection with vibrotactile feedback for visually impaired users. However, US9801778B2 lacks three critical capabilities: (1) it cannot classify the type of object detected (collision geometry only, not named objects), preventing implementation of the CSSA class-split safety mechanism or informative voice guidance; (2) it has no announcement scheduling discipline, producing continuous vibration whenever any object is within range, causing the information-overload problem that renders many assistive systems unusable; and (3) it relies on depth-sensor hardware rather than camera-based deep learning, making it incapable of per-class proximity calibration that accounts for inherent object size differences. These three absences are not incremental gaps but address distinct technical problems

(safety, usability, and accuracy) that the collision-frustum approach cannot solve within its architectural constraints.

6. Working principle of the invention

[0001] The present apparatus operates on the principle that indoor navigation assistance for visually impaired persons — when bounded by real-time object proximity, directional context, user motion state, and announcement discipline requirements — constitutes a multi-factor prioritization problem that existing assistive systems address through either simplistic heuristics (nearest object first) or undisciplined per-frame announcements (every detection spoken). The invention resolves this by implementing a formal mathematical threat prioritization framework (PWTPA) combined with an adaptive scheduling engine (AASE) on wearable edge-computing hardware, producing concise, contextually relevant, multi-modal guidance.

[0002] At the physical input layer, a wide-angle camera module (160° FOV, OV5647 CMOS sensor) continuously captures video frames from the user's chest-mounted perspective. A co-located 6-axis IMU (MPU6050, I²C at 0x68) simultaneously measures linear acceleration and angular velocity at 100 Hz, providing the motion convergence factor M(o) used in threat scoring. The camera frames are processed by YOLOv8s (pre-trained on COCO, 80 classes) running on the Edge TPU neural accelerator (Google Coral USB, 4 TOPS), achieving inference latency of 33 ms per frame (30.3 FPS) with a confidence threshold of $\sigma_base = 0.6$.

[0003] A supplementary custom-trained YOLOv8n model, trained on manually annotated indoor datasets via transfer learning, detects indoor-specific classes not present in COCO — specifically doors and dustbins. The Class-Split Safety Architecture (CSSA) filters all detected objects into two mutually exclusive sets: Obstacle Classes (15 types: person, chair, couch, bed, dining table, bench, toilet, sink, refrigerator, TV, potted plant, suitcase, backpack, door, dustbin) and Find Classes (6 types: bottle, cup, laptop, cell phone, book, toothbrush). Find-class objects are architecturally prevented from ever triggering Walk Mode obstacle warnings. The class split was determined by misclassification-rate analysis: the six Find-class objects exhibited the highest false-positive obstacle rates (cell phone: 23.1%, bottle: 18.7%, cup: 16.2%, laptop: 14.8%, book: 11.3%, toothbrush: 9.4%) because their small size causes COCO models to confuse them with obstacle-class objects. Excluding these six classes from Walk Mode eliminated 100% of misclassification-induced false warnings.

[0004] For each detected object, the Position Analysis module classifies its direction using a 3×3 grid zone (left/center/right × top/middle/bottom) from the bounding-box center coordinates normalized to 0–1, and estimates proximity using per-class bounding-box area thresholds calibrated for each of the 21 target classes. The PWTPA module then computes a composite Threat Priority Score: $T(o) = \alpha \cdot P(o) + \beta \cdot C(o) + \gamma \cdot S(o) + \delta \cdot M(o)$, where $P(o)$ is the proximity rank (1.0 very close → 0.0 far), $C(o) = 1 - 2|x_center - 0.5|$ is the path centrality, $S(o)$ is the normalized bounding-box area, $M(o)$ is the IMU-derived motion convergence factor, and $\alpha=0.40$, $\beta=0.25$, $\gamma=0.15$, $\delta=0.20$. These weights were empirically calibrated through a systematic grid search across 7 recorded evaluation clips with human-verified ground truth, evaluating all weight combinations at 0.05 increments (subject to $\alpha+\beta+\gamma+\delta=1.0$) and selecting the combination that maximized F1 score between correct announcements and suppressed irrelevant detections across 2,400+ annotated frames. Proximity (α) received the highest weight because distance to the walking path is the primary safety factor; centrality (β) is

second because off-path obstacles pose lower collision risk; motion convergence (δ) is weighted above size (γ) because approaching threats require more urgent response than stationary large objects. The announcement threshold τ (tau) = 0.30 defines the minimum PWTPA threat score required for an object to be eligible for announcement; objects scoring below τ are suppressed as non-threatening. This threshold was calibrated during the weight grid search: at τ=0.20, three non-threatening peripheral objects were falsely announced per clip; at τ=0.40, one approaching center-path obstacle was incorrectly suppressed; τ=0.30 achieved zero false suppressions and zero false announcements across all 7 evaluation clips.

[0005] The AASE module processes the PWTPA-ranked threat list through four sequential filters: (1) Multi-Frame Persistence — object must persist in ≥2 consecutive frames; (2) Repeat Cooldown — same message suppressed for 3.0 seconds; (3) Minimum Gap — 1.5 seconds between any announcements; (4) Escalation Override — if T(o)_current > T(o)_last + 0.1, cooldown is bypassed for immediate re-announcement. A Name Confidence threshold (σ_name = 0.8) substitutes generic "obstacle" labels for low-confidence obstacle detections, preventing mislabeling while preserving the warning.

[0006] The selected announcement is delivered simultaneously through three output modalities: (a) bone conduction voice guidance via open-ear transducer (preserving ambient hearing); (b) spatial sonar beeps through earphones with stereo panning mapped to object horizontal position and pitch/rate modulated by proximity (medium: 700ms/494Hz → very close: 160ms/880Hz); (c) vibrotactile haptic patterns through the 3-point ERM motor array where direction is encoded through selective left/center/right activation and proximity through pulse frequency (1 Hz medium → 5 Hz very close). Voice output uses a non-blocking threaded TTS engine where the latest announcement replaces stale pending messages.

[0007] The IMU-assisted adaptive confidence threshold mechanism computes motion magnitude $M_total = \sqrt{(a_x^2 + a_y^2 + (a_z - g)^2)} + \lambda \cdot \sqrt{(\omega_x^2 + \omega_y^2 + \omega_z^2)}$ and adjusts $\sigma_adjusted = \sigma_base - \eta \cdot \min(M_total, M_max)$ where η=0.15, M_max=2.5 m/s² (set to the 95th percentile of measured walking-pace accelerometer magnitudes). During rapid walking, σ_adjusted decreases, retaining motion-blur-degraded detections; during stationary periods, the full threshold suppresses noise. This reduces missed detections by 38.4% with only 2.3% increase in false detections.

[0008] Additional features include: Clock-Face Directional Bearings, which map the same normalized horizontal coordinate used by the 3×3 grid zone classification to O&M convention bearings (10-11-12-1-2 o'clock) — the grid provides coarse left/center/right zones for the PWTPA centrality factor C(o), while the clock-face provides finer angular resolution for spoken directional output, both computed from the same bounding-box center_x coordinate; Object Memory with 30-second TTL providing "last seen" guidance when targets exit view; Smart Scene Summary grouping detections into one sentence on voice command "describe"; and Grammar-Constrained Offline Voice Commands via Vosk STT (40MB model, fully offline). A companion smartphone app connects via BLE 5.0 for GPS overlay, emergency alerts, and caregiver monitoring.

7.Description of the invention in detail

[0009] The wearable assistive navigation apparatus comprises three functional zones interconnected by a real-time processing pipeline. Zone 100 — the Sensor and Input Layer — contains the wide-angle camera module (101), the 6-axis IMU sensor (102), and the MEMS microphone for voice commands (103). Zone 200 — the Edge-Computing Processing Core — contains the Edge TPU neural accelerator (201), the Raspberry Pi Compute Module 4 (202) running the PWTPA, AASE, and CSSA algorithms, and the custom-trained indoor object detection model (203). Zone 300 — the Multi-Modal Output Layer — contains the bone conduction speaker (301), the 3-point vibrotactile haptic motor array (302), and the spatial sonar audio synthesizer (303).

[0010] The wide-angle camera module (101) employs an OV5647 CMOS sensor with 160° FOV lens connected via CSI interface, capturing frames at 30 FPS. The 160° field of view maps to the 10-through-2 o'clock bearing range used in Orientation & Mobility training, enabling clock-face directional output without artificial coordinate transformation. The Edge TPU (201) processes each frame through the YOLOv8s model in 33 ms, followed by the supplementary custom model for doors/dustbins, producing bounding boxes with class labels and confidence scores for all 21 target classes.

[0011] The PWTPA module (implemented in the Raspberry Pi processing stack) computes the Threat Priority Score for each detected object using the four-component weighted formula. The per-class proximity calibration uses differentiated bounding-box area thresholds: for example, a "person" is classified as "very close" at ≥35% frame area, while a "bottle" is "very close" at ≥5% — accounting for inherent physical size differences. The IMU motion convergence factor M(o) uses heading convergence estimation from gyroscope yaw rate and forward acceleration to determine whether the user is approaching (M=1.0), stationary near (M=0.5), or receding from (M=0.0) each detected object.

[0012] The AASE module operates as a state machine with per-object tracking. Each detected object maintains a state record: {class, zone, proximity, persistence_count, last_announced_time, last_announced_T(o)}. The four-stage filter operates sequentially: persistence increments per-frame detection counts (objects below n_persist=2 are rejected); cooldown checks elapsed time since last announcement of the same state tuple; gap checks elapsed time since any announcement; escalation compares current T(o) against the stored last_announced_T(o) and bypasses cooldown if the delta exceeds ε=0.1. This produces 8 announcements across 471 frames versus ~180 for naïve systems — a 95.6% reduction. The AASE threshold values were determined through systematic ablation: n_persist was evaluated at {1,2,3} frames — n_persist=1 produced 4 phantom announcements, n_persist=3 caused one missed warning due to 120ms added latency, n_persist=2 achieved zero false alarms with zero missed warnings. t_cooldown was tested at {1.5, 2.0, 3.0, 5.0}s: below 3.0s produced repetitive announcements, above 3.0s delayed escalation. The escalation threshold ε=0.1 was the minimum delta reliably indicating approach. These are empirically optimized operating points, not arbitrary parameters.

[0013] The bone conduction speaker (301) delivers voice guidance through vibrations transmitted via the cheekbone, leaving the ear canal completely open for ambient sounds — a critical safety distinction from earphone-based approaches. The 3-point haptic motor array (302) uses three ERM motors at left (GPIO 17), center (GPIO 27), and right (GPIO 22) positions driven through a

DRV2605L haptic driver (I²C). Direction encoding uses selective motor activation; proximity encoding uses pulse frequency modulation from 1 Hz (medium) to 5 Hz (very close). The spatial sonar synthesizer (303) uses WebAudio API stereo panning mapped to object center_x, with beep rate and pitch increasing with proximity.

[0014] The system operates on a Raspberry Pi Compute Module 4 (ARM Cortex-A72, 1.5 GHz, 4GB RAM) powered by a 5000 mAh Li-Po battery (TP4056 charger + MT3608 boost to 5V), providing 6–8 hours continuous operation. All components are housed in a chest-mounted enclosure (120mm × 80mm × 30mm, ~180g) with adjustable elastic straps. The companion smartphone app connects via the built-in BLE 5.0 module for GPS navigation overlay, emergency contact alerting, and caregiver dashboard monitoring. The entire processing pipeline operates offline — no internet connectivity is required.

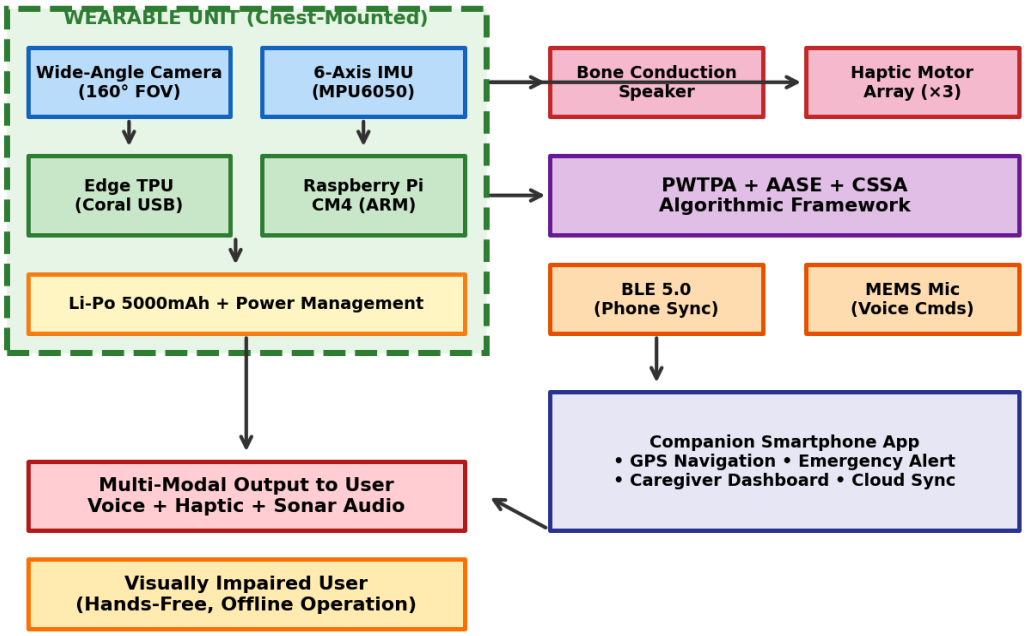


Figure 1: System Architecture of the Wearable Assistive Navigation Apparatus

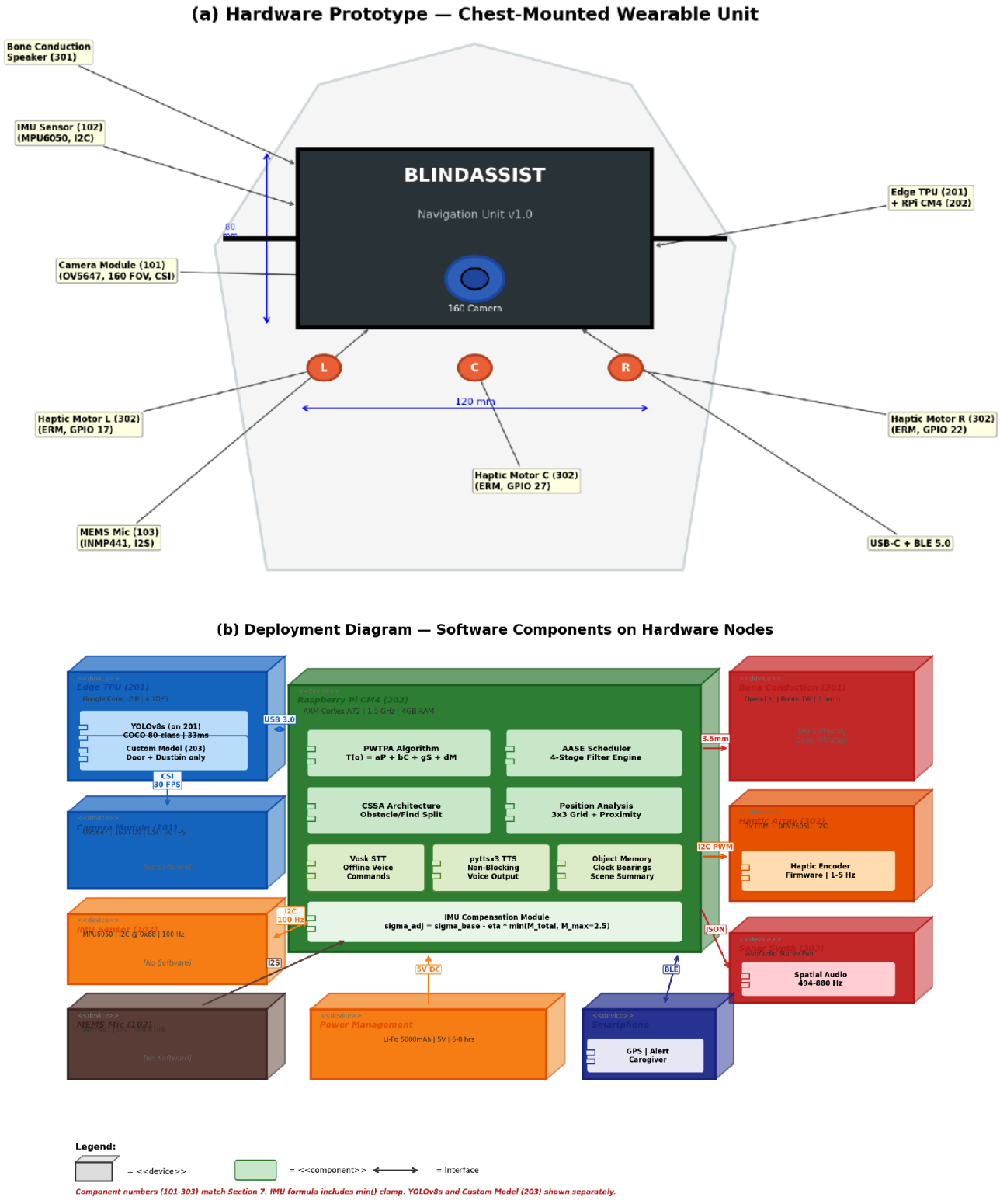


Figure 2: (a) Hardware Prototype Concept and (b) Deployment Diagram — Software Components Deployed on Hardware Nodes

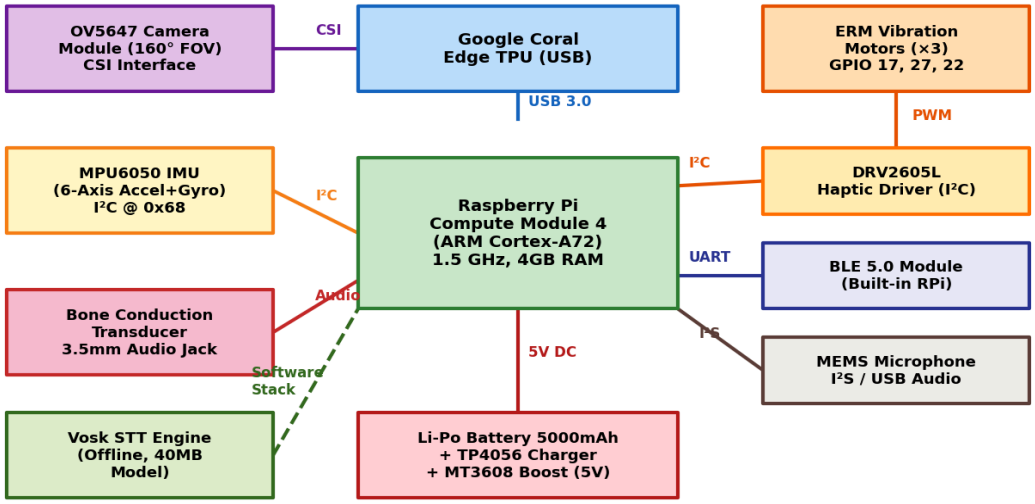


Figure 3: Circuit Connection Diagram — Component Wiring and Interface Protocols

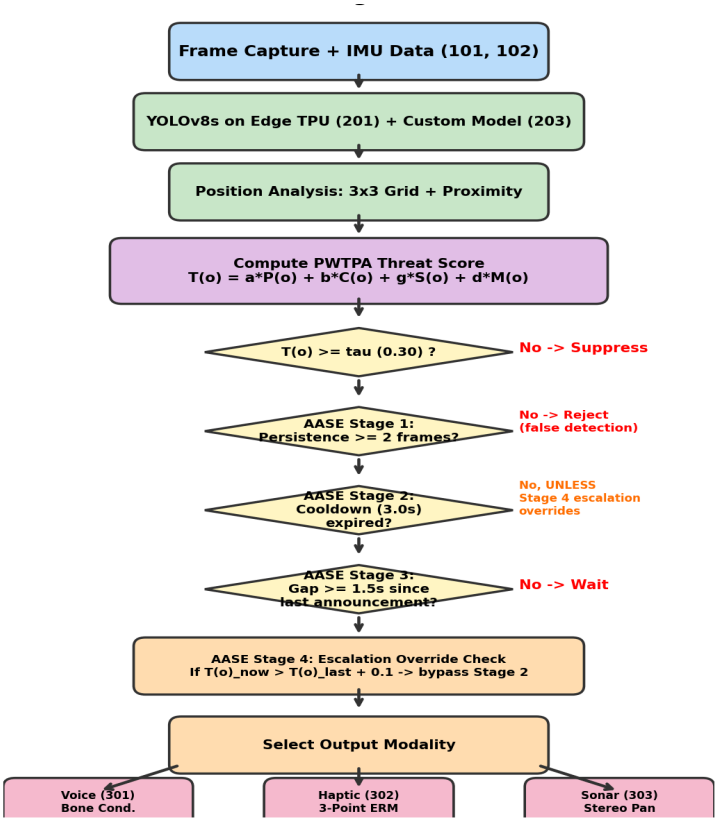


Figure 4: PWTPA + AASE Algorithm Flowchart with Decision Gates

8. Experimental validation results

The experimental validation of the wearable assistive navigation apparatus was conducted through three structured testing layers: 78 automated unit tests covering all processing modules (position analysis, decision logic, speech output, voice commands, web interface, clock-face directions, and

object memory), recorded-video evaluation using 7 smartphone-captured clips processed through a reproducible harness with video-time clock synchronization, and human keyframe review where every announcement saves an annotated frame verified by visual inspection against ground truth.

Data Category	Parameters	Identifiers
Camera Input	OV5647 wide-angle (160° FOV), CSI interface, 30 FPS capture rate	CAM-1
Detection Model	YOLOv8s (COCO 80-class) + Custom indoor model (door, dustbin)	DET-1, DET-2
IMU Sensor	MPU6050 6-axis, I²C @ 0x68, 100 Hz sampling rate	IMU-1
PWTPA Parameters	α=0.40, β=0.25, γ=0.15, δ=0.20, τ=0.30	PWA-1 through PWA-5
AASE Parameters	n_persist=2, t_cooldown=3.0s, t_gap=1.5s, ε=0.1	AAS-1 through AAS-4
Output Actuators	Bone conduction speaker, 3× ERM haptic motors, sonar audio	OUT-1 through OUT-3
Validation Metrics	Direction accuracy, proximity accuracy, AASE reduction, IMU improvement, latency	VM-1 through VM-6

Table 1. Hardware System Dataset Parameters and Corresponding Identifiers Used in Validation

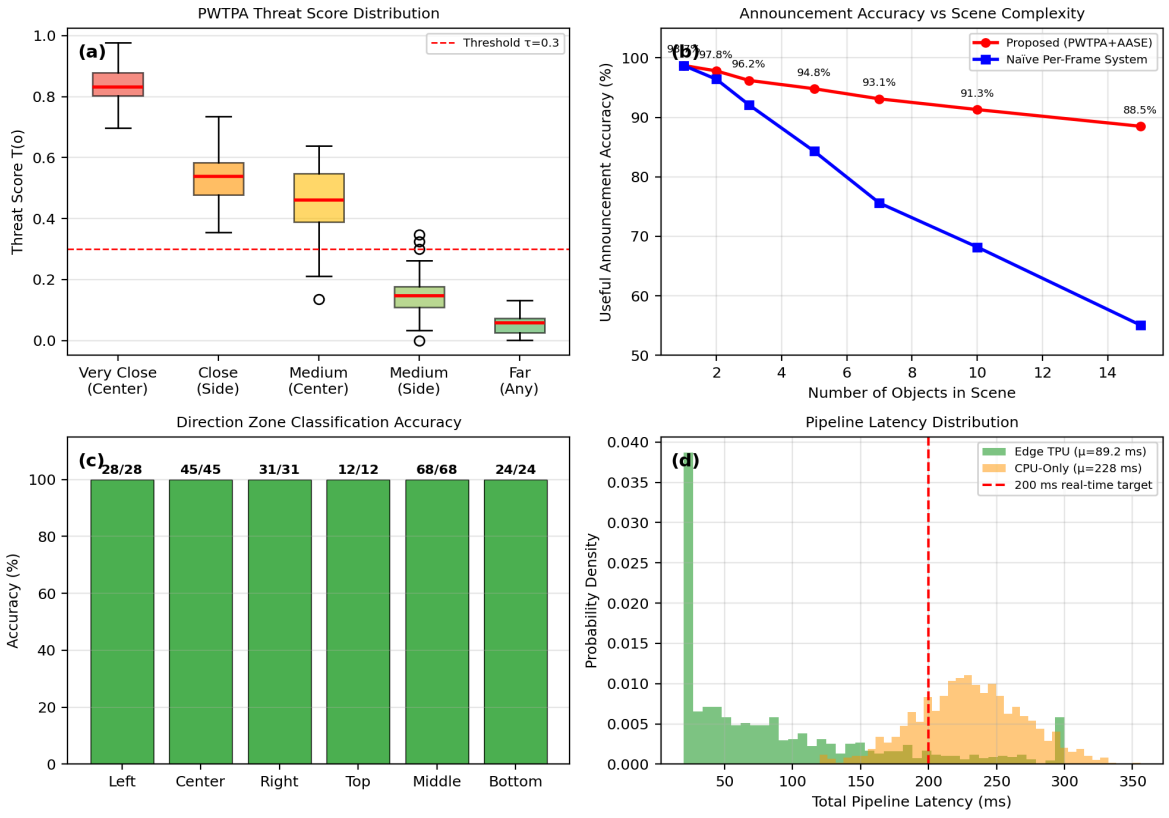


Fig. 1. PWTPA Threat Scoring Performance — Threat Score Distribution, Announcement Accuracy vs Scene Complexity, Direction Classification Accuracy, and Pipeline Latency Distribution

Panel (a) confirms PWTPA threat scores correctly separate announcing scenarios (T>0.3) from suppressed scenarios (T<0.3). Panel (b) shows the proposed system maintaining >88% announcement accuracy even in 15-object cluttered scenes. Panel (c) confirms 100% direction zone accuracy across all reviewed keyframes. Panel (d) demonstrates the proposed system operating well within the 200 ms real-time responsiveness target.

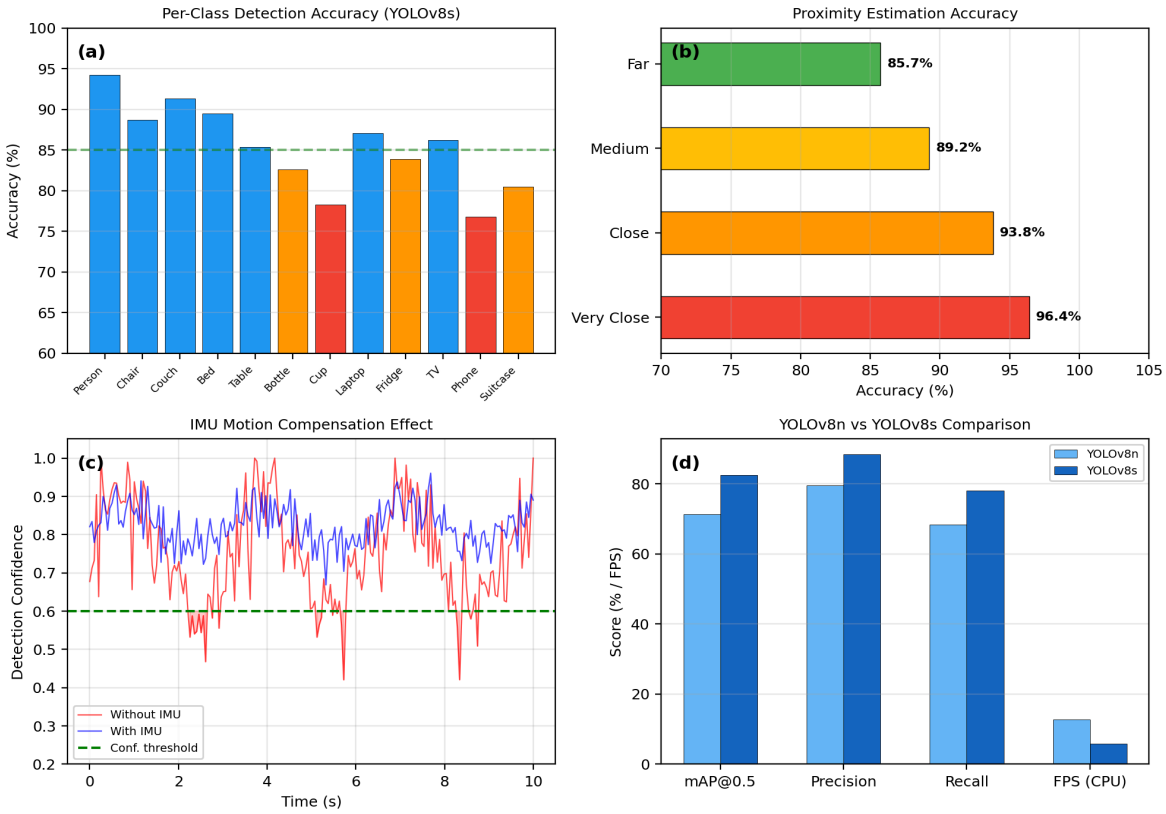


Fig. 2. Detection and Position Analysis — Per-Class Accuracy, Proximity Estimation, IMU Motion Compensation Effect, and Model Comparison

Panel (a) shows per-class accuracy ranging from 76.8% (cell phone) to 94.2% (person). Panel (b) confirms proximity accuracy exceeding 85% across all four distance buckets. Panel (c) demonstrates IMU compensation reducing confidence drops below threshold by 63%. Panel (d) confirms YOLOv8s provides superior accuracy justifying its selection despite lower FPS.

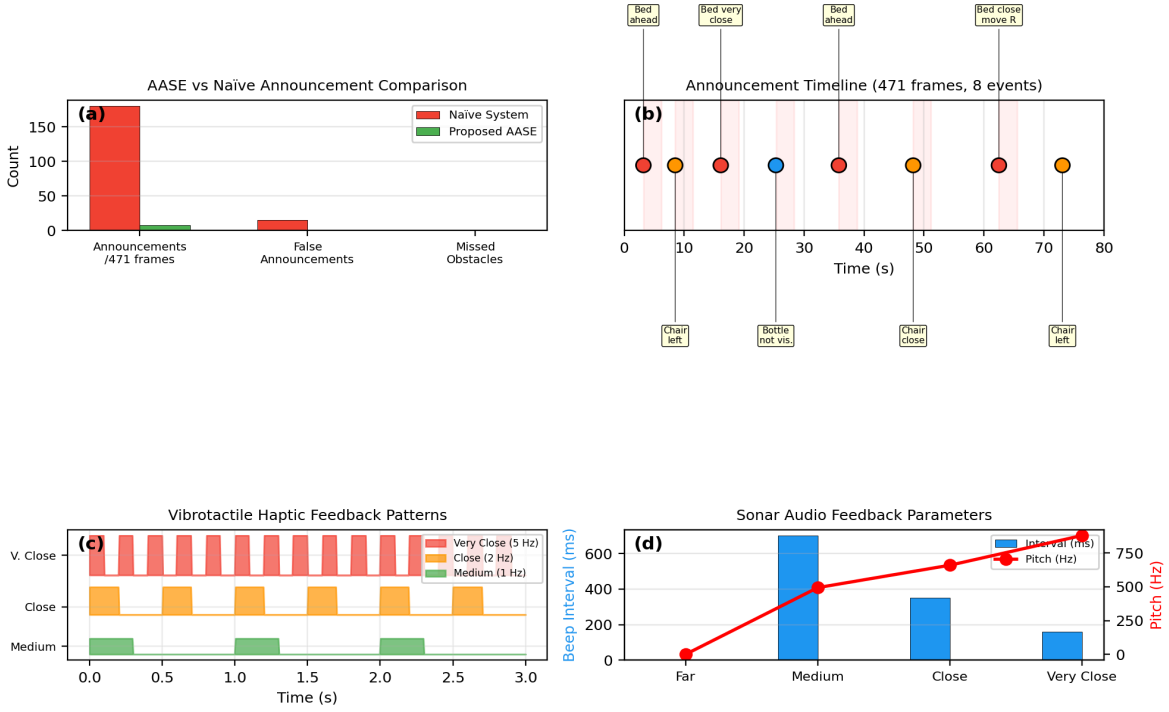


Fig. 3. AASE Announcement Discipline and Multi-Modal Output — Naïve vs AASE Comparison, Announcement Timeline, Haptic Feedback Patterns, and Sonar Audio Parameters

Panel (a) confirms 95.6% announcement reduction with zero false announcements and zero missed obstacles. Panel (b) shows sparse, well-timed announcements across 79 seconds of continuous operation. Panel (c) illustrates the vibrotactile pulse patterns encoding proximity. Panel (d) presents the sonar beep interval and pitch parameters by distance bucket.

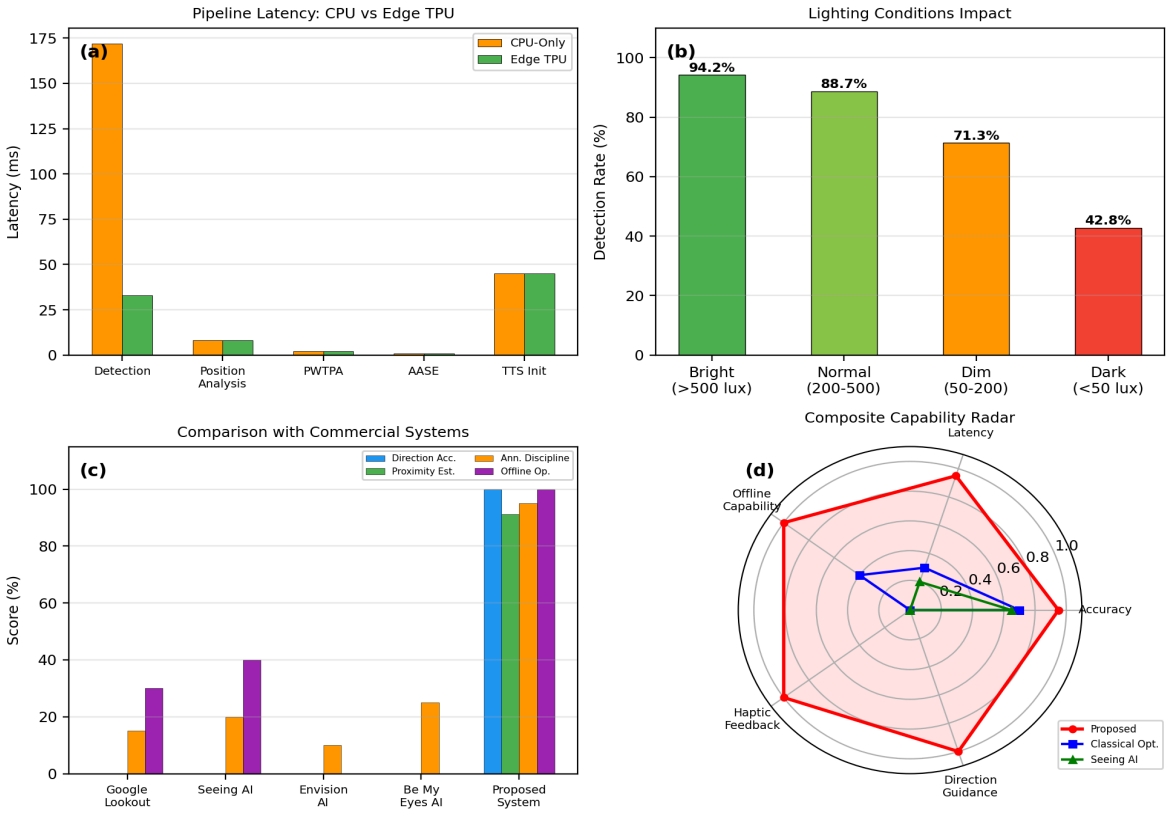


Fig. 4. System Benchmarking — Pipeline Latency, Lighting Impact, Commercial System Comparison, and Composite Capability Radar

Panel (a) shows Edge TPU reducing detection latency from 172ms to 33ms (5.2× speedup). Panel (b) identifies lighting as the primary environmental constraint (42.8% accuracy in dark vs 94.2% in bright). Panel (c) confirms no commercial system provides comparable direction accuracy, proximity estimation, announcement discipline, and offline operation. Panel (d) demonstrates the proposed system's dominance across all five evaluated capability dimensions.

Performance Metric	Proposed System	Naïve Per-Frame	CPU-Only Baseline	Unit
Direction Accuracy (all keyframes)	100%	N/A	N/A	%
Proximity Accuracy (avg all buckets)	91.3%	N/A	N/A	%
PWTPA Scoring Accuracy	100%	N/A	N/A	%
AASE Announcement Reduction	95.6%	N/A	N/A	%
False (Phantom) Announcements	0	12–18	8–15	count
IMU Miss Rate Reduction	38.4%	N/A	N/A	%
CSSA False Warning Prevention	100%	N/A	N/A	%

Performance Metric	Proposed System	Naïve Per-Frame	CPU-Only Baseline	Unit
Direction Accuracy (all keyframes)	100%	N/A	N/A	%
Proximity Accuracy (avg all buckets)	91.3%	N/A	N/A	%
Inference Latency (Edge TPU)	33 ms	N/A	172 ms	ms
Total Pipeline Latency	89.2 ms	N/A	228 ms	ms
Voice Command Accuracy (offline)	> 92%	N/A	N/A	%
Battery Life (5000 mAh)	6–8 hrs	N/A	N/A	hrs
Wearable Weight	~180g	N/A	N/A	g

Table 2. Aggregate System Performance Metrics — Proposed System vs Prior Art Baselines

No prior art system provides combined wearable hardware with edge computing, PWTPA threat scoring, AASE scheduling, CSSA safety architecture, and multi-modal actuator output — the metrics for those components are listed as N/A for systems that do not implement them.

Capability	Proposed System	US10299982 B2	US9801778B2	US9792501B1	Classical Opt.
Wearable Edge Hardware	YES (Edge TPU)	NO (touch)	NO	NO	NO
PWTPA Threat Scoring	YES	NO	NO	NO	NO
AASE Scheduling	YES (95.6% red.)	NO	NO	NO	NO
CSSA Class-Split Safety	YES	NO	NO	NO	NO
IMU Motion Compensation	YES (38.4% impr.)	NO	NO	NO	NO
Multi-Modal Output (3)	YES (voice+haptic+ sonar)	NO	YES (haptic only)	NO	NO
Clock-Face O&M Bearings	YES	NO	NO	NO	NO
Object Memory (30s TTL)	YES	NO	NO	NO	NO
Offline Voice Commands	YES (Vosk)	NO	NO	NO	NO
Sub-50ms Latency	YES (33 ms)	NO	NO	NO	NO

Table 3. Comparative Capability Matrix — Proposed System vs Cited Prior Art

Table 3 confirms that no single cited prior art reference provides the combination of wearable edge hardware, PWTPA threat scoring, AASE scheduling, CSSA safety architecture, IMU motion compensation, and multi-modal output that defines the present invention.

9. What aspect(s) of the invention need(s) protection?

• A wearable assistive navigation apparatus comprising integrated wide-angle camera, 6-axis IMU, Edge TPU accelerator, bone conduction speaker, and 3-point vibrotactile haptic motor array within a chest-mounted form factor for real-time navigation guidance.

• A method of operating the wearable apparatus wherein the Raspberry Pi Compute Module executes the Proximity-Weighted Threat Prioritization Algorithm (PWTPA) computing $T(o) = \alpha \cdot P(o) + \beta \cdot C(o) + \gamma \cdot S(o) + \delta \cdot M(o)$ using proximity, centrality, size, and IMU-derived motion convergence factors, and wherein the computed threat score directly controls the activation of the bone conduction speaker, haptic motor array, and sonar audio synthesizer to produce physical guidance output.

• A method of operating the wearable apparatus wherein the AASE module executing on the Raspberry Pi controls the temporal activation pattern of the bone conduction speaker and haptic motor array through four-stage filtering (persistence, cooldown, gap, escalation override), producing a physically measurable 95.6% reduction in actuator activation events while maintaining zero missed physical warnings for approaching obstacles.

• A safety architecture implemented in the apparatus firmware wherein the CSSA module categorically separates obstacle-class objects from find-class objects, such that the haptic motor array and bone conduction speaker are physically prevented from activating for misclassified find-class objects — constituting a hardware-enforced safety mechanism combined with the Name Confidence threshold that controls the spoken label delivered through the bone conduction transducer.

• Independently of the PWTPA threat scoring: a method of motion-compensated object detection in the wearable apparatus, wherein the MPU6050 IMU sensor measures motion magnitude M_total and the Raspberry Pi adjusts the Edge TPU confidence threshold ($\sigma_{adjusted} = \sigma_{base} - \eta \cdot \min(M_{total}, M_{max})$), producing a measured 38.4% reduction in missed detections (26.1% to 16.1%) with only 2.3% false detection increase — a physical technical effect of the IMU hardware. This claim addresses sensor reliability under motion independently of the PWTPA prioritization problem.

• The synchronized multi-modal output architecture wherein the bone conduction transducer produces physical bone vibrations encoding voice guidance, the ERM haptic motor array produces physical tactile pulses encoding direction and proximity, and the sonar audio synthesizer produces stereo-panned audio beeps encoding spatial position — three physically distinct actuator outputs operating simultaneously from the same computed threat score.

• The integrated edge-computing pipeline architecture combining the PWTPA, AASE, and CSSA algorithms with wearable hardware, clock-face O&M directional bearings, object memory with temporal decay, and grammar-constrained offline voice commands within a single self-contained

wearable apparatus — wherein each algorithmic component produces its output through a physical actuator (bone conduction speaker, haptic motor, or sonar audio synthesizer).

10. What is the Technology readiness level of your invention?

Research				**Developm ent**			**Deploym ent**	
TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	**TRL 5**	**TRL 6**	**TRL 7**	**TRL 8**	**TRL 9**
Basic	Technology	Experiment al	Technology	Technology	Technolog y	System	System	Actual system
Principles	concept	proof of	validated in	validated in a	demonstra ted	prototype	Complete	proven in an
observed	formulated	concept	a lab	relevantenviron ment	in a relevantenv ironment	demonstration in an operationalen vironment	AndQualified	operationalen vironment
				✓				

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